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Paleomagnetism and $^{40}Ar/^{39}Ar$ dating of the early Cretaceous Florianópolis dike swarm (Santa Catarina Island), Southern Brazil

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Abstract

The Florianópolis (FL) basaltic dikes are well exposed on Santa Catarina Island, close to the present southeastern border of the Paraná Magmatic Province in Southern Brazil, and their intrusion was related to the tectonic and magmatic evolution of rifting in the southern Atlantic. The dikes are vertical or subvertical, and cut the crystalline basement rocks, mainly Late Proterozoic granites. Most of the dikes exhibit NE trends, corresponding to structural trends of the crystalline basement. A few NW-trending dikes can also be observed, occasionally crosscutting the NE dikes. Samples from 73 dikes were collected for paleomagnetic work, and subjected to both alternating field and thermal demagnetization. Characteristic magnetizations of normal and reversed polarities interpreted as original thermoremanences are carried by Ti-poor magnetites showing remanent coercive forces in the range 18–45 mT. The corresponding paleomagnetic pole (FL pole) is located at 3.3°E, 89.1°S ($N = 65$; $\alpha_{95} = 2.6°$; $k = 47$). Samples for $^{40}Ar/^{39}Ar$ dating were collected at the same paleomagnetic sites. Plagioclase samples from nine dikes were analyzed by stepwise heating; the apparent age spectra are strongly discordant due to excess ^{40}Ar contamination, but five samples yielded plateau ages ranging from 119.0 ± 0.9 Ma to 128.3 ± 0.5 Ma. One additional sample yielded an isochron age of 121.5 ± 0.5 Ma. The inferred ages apparently define two modes at $\sim$119–122 Ma and $\sim$126–128 Ma, although the data set is probably insufficient to determine whether dike intrusion was continuous or episodic. The ages of 128–119 Ma are coeval with the final stages of the rifting at this latitude, suggesting that the emplacement of the FL dikes is related to extension of the continental crust just prior to the formation of oceanic crust. By comparing the FL pole with the existing paleomagnetic poles for the NW Ponta Grossa dike swarm (mainly 130.5 Ma) to the north, the Paraná volcanics (mainly 133–132 Ma), and the Central Alkaline Province (127–130 Ma) on the western side of the Paraná Province it is concluded that the differences in pole positions reflect plate displacements, and therefore the FL pole is probably dominated by the dikes with ages younger than 127 Ma. © 1998 Elsevier Science B.V. All rights reserved.

Keywords: Paleomagnetism; Geochronology; Mesozoic; Dike swarm; Paraná Basin

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1. Introduction

The apparent polar wander path (APWP) for South America is still poorly defined for most geologic time intervals. Several well-dated Mesozoic reference poles have been established recently, mainly for the Early Cretaceous (Ernesto et al., 1990; Raposo and Ernesto, 1995; Ernesto et al., 1996) with ages covering a very narrow interval, mainly 133 to 127 (Renne et al., 1992a, 1996a; Velázquez et al., 1992), and Late Cretaceous (Butler et al., 1991; Montes-Lauar et al., 1995) with ages in the 90–70 Ma interval. As pointed out by Somoza (1994) there is a lack of reliable paleomagnetic poles with ages between those two epochs, and the only pole that can be used as a reference paleopole (Cerro Barcino pole, Somoza, 1994) still shows uncertainties regarding age and origin of the remanent magnetization of the Cerro Barcino sediments.

The South American APWP for the Cretaceous is particularly important to be investigated in detail since it may help to elucidate the early history of the separation between South America and Africa prior to effective ocean floor formation, when no magnetic anomalies nor other oceanic features are available for reconstructing continental drift. Even during the drifting stage the tracking of plate motion is hampered by the presence of the magnetic quiet zone from about 120 to 83.5 Ma (Gradstein et al., 1994).

The opening of the South Atlantic ocean was preceded by intense magmatism in Southeastern Brazil (Fig. 1), expressed by the voluminous continental flood basalts of the Paraná Basin (Piccirillo and Melfi, 1988), and the related intrusive tholeiitic magmatism represented by the NW–SE trending Ponta Grossa dike swarm (Raposo and Ernesto, 1995; Renne et al., 1996a; Piccirillo et al., 1990), as well as NE–SW trending swarms such as those occurring on Santa Catarina Island (Florianópolis dikes) and the coastal area from Santos to Rio de Janeiro (Almeida, 1986; Comin-Chiaramonti et al., 1983). The ages of the Paraná basalts are well constrained by $^{40}Ar/^{39}Ar$ dating indicating a very narrow interval (roughly 133–132 Ma) for the main magmatic activity (Renne et al., 1992a,b; Turner et al., 1994; Stewart et al., 1996). The Ponta Grossa dikes were also injected during a brief interval, mainly 131.3 ± 0.5 to 129.2 ± 0.5 Ma (Renne et al., 1996a), with minor subsequent intrusion as late as 120 Ma. For the Santos-Rio de Janeiro swarm ages are in the range of 132–119 Ma (Turner et al., 1994; Renne et al., 1993). These data indicate a tendency for younger ages concentrating towards the continental margin, and reflecting extension related to the final stages of rifting. According to Austin and Uchupi (1982) the South Atlantic rift progressed from south to north, reaching latitudes north of 30°S (southern Paraná Basin) by about 113 Ma. At these latitudes the continental crust was highly extended between 130 and 120 Ma (Chang et al., 1992). Therefore, if the Florianópolis dikes can be linked to this stretching event, they should provide information about rifting prior to the formation of oceanic crust.

This paper presents paleomagnetic and $^{40}Ar/^{39}Ar$ data from the Florianópolis dike swarm in order to contribute to a better definition of the APWP segment for the Cretaceous of South America, and to constrain the early opening history of the South Atlantic Ocean.

2. Geological setting

Dikes of the Florianópolis swarm occur on Santa Catarina Island, in and around Florianópolis city, as shown in Fig. 1. Although references in the literature are rare, these dikes have been presumed to be related to the Early Cretaceous Serra Geral magmatism of the Paraná Basin (Bortoluzzi et al., 1987).

The Florianópolis dikes are generally well exposed. They cut the crystalline basement rocks, mainly the post-tectonic Upper Proterozoic Pedras Grandes intrusive granitic suite (Fig. 1) as defined by Silva (1987). Dike thicknesses vary from 0.1 to 70 m although values between 0.5 and 10 m are most frequent. Thicker dikes, in which chilled margins are present, show variation in grain size. The majority of the sampled dikes (52 dikes) trend N20°E to N65°E (Table 1) paralleling the main structures of the crystalline basement, and few dikes (8 dikes) trend N10°W to N40°W. In the Joaquina beach (site 130 in Fig. 1) the NW system is seen cutting NE dikes, indicating that the former postdates the main system. All dikes are vertical or subvertical.

Preliminary geochemical data (Marques et al., 1993) indicate that the FL dikes are mainly basalts,

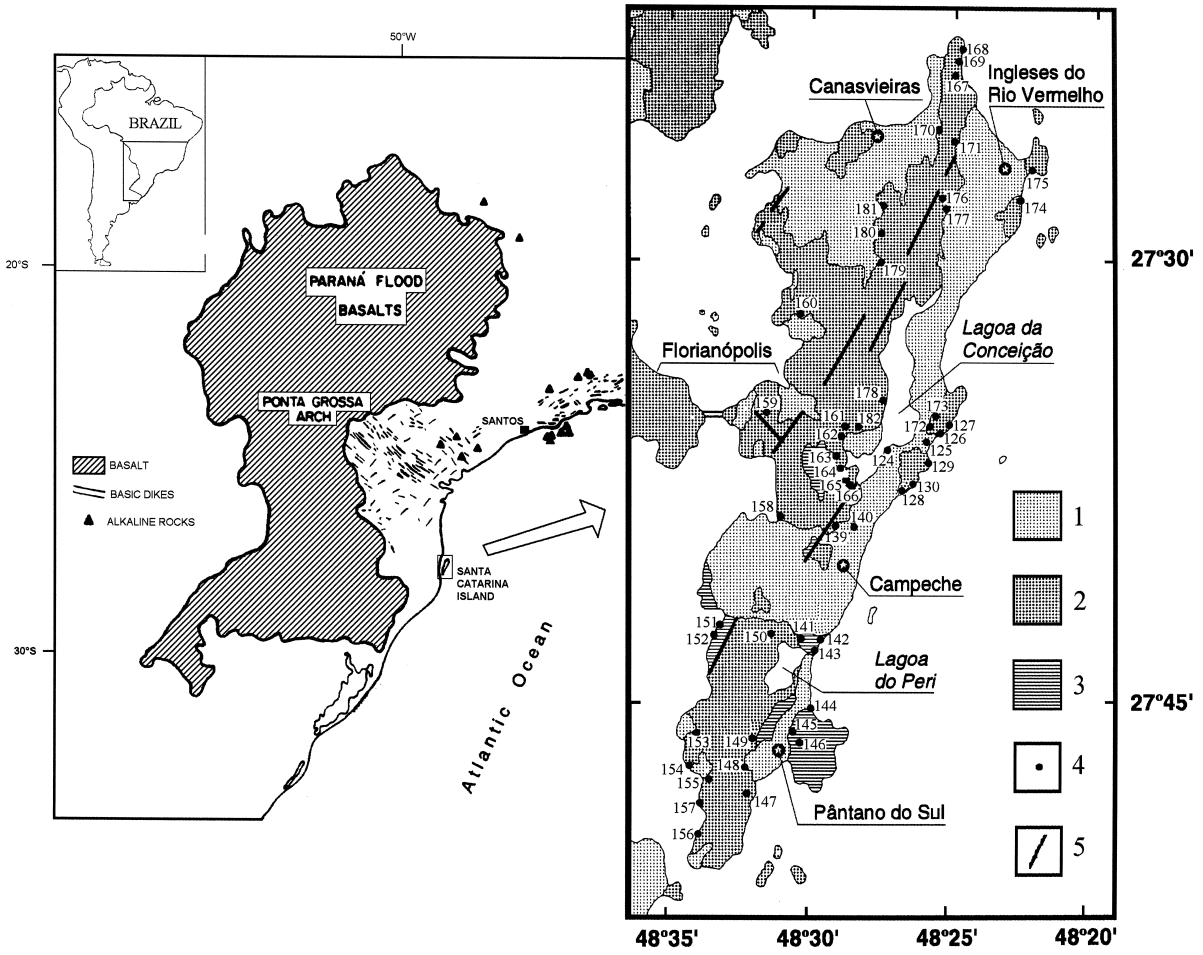


Fig. 1. Distribution of Mesozoic magmatism in South Brazil (left), modified from Chang et al. (1992), and generalized geological map of Santa Catarina Island (right): 1) Cenozoic sediments; 2) Proterozoic Pedras Grandes Granitic Suite; 3) Eopaleozoic Campo Alegre Formation (volcanic-sedimentary sequence); 4) Sampling sites; 5) Faults and/or fractures.

and no significant differences were observed between the groups with different orientations. The dikes are similar geochemically to the Paraná flood basalts (Serra Geral Formation, Piccirillo and Melfi, 1988; Marques et al., 1989) and the diabases from the Ponta Grossa dike swarm (Piccirillo et al., 1990).

3. Paleomagnetic analyses and results

For the paleomagnetic work samples were collected from 71 dikes outcropping on beaches and roadcuts on Santa Catarina Island (Fig. 1), and from two dikes (sites 183 and 184, Table 1) on São Francisco Island, northward of the main area (Fig. 1). These two dikes are very similar to the Florianópolis dikes, and cut the same granitic rocks. In some sampling points more than one dike was sampled, and they are discriminated by letters following the corresponding outcrop number (Table 1). As the outcrops were highly fractured most samples are hand-blocks but cylinders were also collected with the aid of a gasoline-powered rock drill. In general, 3 blocks or 6 to 9 cylinders were collected from each dike (site) distributed through the exposed

Table 1
Paleomagnetic results of the Florianópolis dikes

Site	Coordinates				Mean magnetization direction						Virtual geomagnetic poles			
	Long. (°W)	Lat. S °S	Strike	Width (m)	N/n	Dec. (°)	Inc. (°)	α_{95} (°)	k	Pol.	Long. (°E)	Lat. (°)	dm (°)	dp (°)
124	48.46	27.60	?	?	no consistent results									
125A	48.43	27.61	N15E	10	8/6	3.7	−51.8	4.7	206.9	N	98.4	84.2	6.4	4.3
125B	48.43	27.61	N15E	2	4/4	13.9	−67.7	10.6	75.7	N	110.6	64.7	17.8	14.9
126	48.52	27.59	N10E	3	no consistent results									
127	48.42	27.59	N30E	15	8/8	197.2	48.1	6.5	74.3	R	231.6	−74.8	8.5	5.5
128A	48.45	27.63	N20E	5	no consistent results									
128B	48.45	27.63	N40W	0.5	3/3	349.3	−75.1	3.3	1361.1	N	140.3	54.9	6.1	5.6
129A*	48.45	27.63	N5E	10	8/8	213.6	−75.6	5.3	109.8	I	326.2	−4.3	9.8	9.0
129B*	48.45	27.63	N10E	10	4/4	301.1	−67.4	7.0	171.7	I	177.6	40.5	11.7	9.8
130A	48.45	27.63	N20E	20	13/13	183.2	43.0	2.2	368.6	R	179.8	−86.1	2.7	1.7
130B	48.45	27.63	N30W	0.6	4/4	4.1	−64.0	4.4	441.8	N	122.5	71.6	7.0	5.6
130C	48.45	27.63	N10W	2	4/4	356.4	−61.0	2.7	1198.7	N	142.1	75.3	4.1	3.1
139	48.48	27.65	N20E	?	4/4	5.2	−45.2	4.6	406.7	N	32.0	85.2	5.8	3.7
140A	48.48	27.65	N30E	60	14/14	172.5	37.6	3.8	108.6	R	83.6	−80.5	4.5	2.7
140B	48.48	27.65	N30E	60	14/13	195.2	48.0	4.4	90.6	R	231.0	−76.6	5.7	3.7
141A	48.51	27.71	N25E	?	4/4	172.9	54.4	4.8	373.6	R	350.0	−80.6	6.7	4.7
141B	48.51	27.71	N20E	?	8/7	352.4	−39.8	3.9	243.2	N	256.2	81.4	4.7	2.8
142A	48.50	27.72	N30E	30	16/13	8.2	−53.9	4.0	108.6	N	87.4	80.2	5.6	3.9
142B	48.50	27.72	N45E	30	12/12	4.8	−58.3	2.4	324.4	N	113.4	78.0	3.6	2.6
142C	48.50	27.72	N45E	0.6	4/4	7.1	−58.0	1.5	3626.2	N	104.8	77.6	2.3	1.7
143A	48.50	27.72	N35E	5	4/4	187.0	38.1	3.3	798.6	R	178.3	−81.0	3.9	2.3
143B	48.50	27.72	N35E	30	7/4	345.4	−35.0	4.7	382.9	N	250.7	74.2	5.4	3.1
143C	48.50	27.72	N30E	50	13/11	350.3	−50.8	4.5	103.5	N	194.8	80.7	6.1	4.1
143D	48.50	27.72	N20E	0.7	4/4	328.4	−27.3	6.3	217.4	N	239.3	57.8	6.8	3.7
144A	48.50	27.75	N45E	30	11/5	168.0	48.9	7.8	98.6	R	25.9	−79.2	10.2	6.7
144B	48.50	27.75	N45E	0.3	4/4	346.9	−21.0	3.0	920.6	N	272.5	69.3	3.2	1.7
145	48.51	27.76	?	?	11/7	175.5	43.0	3.3	339.5	R	75.3	−85.1	4.1	2.5
146	48.51	27.71	?	?	8/8	164.8	47.5	2.6	473.3	R	33.9	−76.6	3.3	2.2
147	48.53	27.79	?	?	8/4	347.7	−35.0	1.9	2244.1	N	255.8	75.9	2.2	1.3
148	48.53	27.78	N30E	?	8/8	14.4	−39.5	2.4	515.7	N	22.3	75.9	2.9	1.8
149	48.53	27.77	N35E	?	8/8	4.0	−41.2	4.9	130.9	N	353.5	84.5	5.9	3.6
150	48.51	27.71	N45E	5	7/7	176.4	40.1	4.6	170.7	R	96.6	−84.1	5.6	3.4
151	48.55	27.70	N55E	?	4/4	354.1	−39.6	5.0	332.9	N	264.7	82.5	6.1	3.6
152	48.55	27.71	N20E	?	4/4	2.0	−50.3	3.0	927.8	N	105.0	86.2	4.1	2.7
153	48.57	27.76	N30E	?	11/8	15.3	−45.0	2.5	511.7	N	39.9	76.4	3.1	2.0
154*	48.56	27.78	?	?	8/4	301.8	−76.4	1.9	2405.0	I	159.8	38.5	3.5	3.2
155	48.56	27.79	?	?	4/4	342.2	−49.3	1.0	7762.4	N	208.5	74.3	1.4	0.9
156	48.56	27.82	N40E	?	15/10	174.0	53.0	1.7	769.8	R	351.7	−82.2	2.4	1.7
157	48.52	27.64	?	?	8/8	344.5	−35.5	3.9	201.0	N	247.7	73.7	4.5	2.6
158	48.52	27.64	N30E	?	6/6	171.9	56.6	4.5	221.0	R	344.8	−78.3	6.5	4.7
159	48.54	27.59	?	?	4/4	178.0	45.2	3.6	655.5	R	67.6	−88.0	4.6	2.9
160	48.51	27.53	N80E	?	9/9	176.9	46.8	3.7	194.8	R	30.6	−87.2	4.8	3.1
161A	48.48	27.60	?	?	12/12	357.9	−47.4	1.7	679.7	N	193.2	87.9	2.2	1.4
161B	48.48	27.60	?	?	8/6	352.0	−52.5	3.3	409.4	N	181.0	81.2	4.6	3.1
162	48.48	27.61	N10E	?	8/8	3.5	−39.0	1.9	848.6	N	341.8	83.6	2.3	1.4
163	48.49	27.61	N20E	?	8/7	347.5	−40.2	2.8	471.3	N	241.2	77.8	3.4	2.0
164A	48.48	27.62	N40E	30	6/4	358.5	−43.4	2.5	1378.3	N	280.4	87.3	3.1	1.9
164B	48.48	27.62	N20E	3	8/8	356.3	−29.5	3.1	321.4	N	294.7	77.7	3.4	1.9
165	48.48	27.62	N45E	1.5	9/9	1.0	−43.7	3.1	274.8	N	335.1	87.7	3.9	2.4
166	48.48	27.63	?	?	no consistent results									
167	48.42	27.40	N20E	15	11/11	5.7	−40.4	3.3	190.6	N	3.0	83.3	4.0	2.4

Table 1 (continued)

Site	Coordinates				Mean magnetization direction						Virtual geomagnetic poles			
	Long. (°W)	Lat. S °S	Strike	Width (m)	N/n	Dec. (°)	Inc. (°)	α_{95} (°)	k	Pol.	Long. (°E)	Lat. (°)	dm (°)	dp (°)
168A	48.41	27.39	N30W	20	6/6	3.8	−36.8	1.3	2496.0	N	339.0	82.3	1.6	0.9
168B	48.41	27.39	N30W	10	4/4	14.8	−48.9	3.1	870.3	N	55.6	76.8	4.1	2.7
169	48.41	27.39	N–S	?	9/9	358.6	−41.1	1.9	720.3	N	292.4	86.0	2.3	1.4
170	48.42	27.43	N30E	1.5	8/8	15.3	−46.6	3.8	215.8	N	47.1	76.4	4.9	3.1
171A	48.41	27.44	N60E	?	12/12	2.2	−31.4	1.7	650.3	N	323.0	79.3	1.9	1.1
171B	48.41	27.44	N20E	?	11/11	359.5	−23.6	1.8	632.9	N	309.8	74.9	1.9	1.0
172	48.43	27.60	N80W	15	8/8	184.5	43.6	2.3	590.4	R	194.2	−85.4	2.8	1.8
173	48.43	27.59	N15E	?	8/8	184.2	46.8	4.3	169.4	R	228.7	−86.3	5.5	3.5
174A	48.37	27.47	N50E	20	12/12	181.1	57.2	3.4	161.0	R	306.8	−79.6	5.0	3.7
174B	48.37	27.47	N50E	20	8/4	176.7	40.6	2.0	2155.7	R	96.3	−84.8	2.4	1.5
174C	48.37	27.47	N60E	5	no consistent results									
174D	48.37	27.47	N50E	50	10/7	173.1	49.6	2.5	608.5	R	14.1	−83.3	3.3	2.2
175	48.37	27.45	N10E	5	8/5	191.5	52.7	9.5	65.5	R	254.8	−78.5	13.1	9.1
176	48.42	27.48	N45E	?	8/8	13.5	−44.3	2.4	517.8	N	37.8	77.9	3.1	1.9
177	48.42	27.48	30E	3	8/7	4.9	−57.6	2.4	657.6	N	111.8	78.5	3.5	2.5
178	48.46	27.42	N60E	?	7/7	347.8	−54.1	2.3	693.8	N	183.8	77.3	3.2	2.3
179	48.46	27.49	N20W	?	8/8	173.0	40.9	3.8	217.8	R	72.6	−82.4	4.6	2.8
180	48.46	27.47	N30E	?	10/10	359.4	−32.5	2.8	305.3	N	308.1	80.2	3.1	1.8
181	48.46	27.47	N40W	4	8/8	12.1	−55.1	4.7	137.4	N	83.1	76.9	6.7	4.8
182	48.48	27.60	N20E	15	9/9	357.4	−43.3	1.6	1009.7	N	266.4	86.7	2.0	1.3
183	48.55	26.21	N30E	?	4/4	1.4	−43.6	3.0	917.8	N	12.0	88.5	3.8	2.4
184	48.51	26.15	N30W	60	15/13	356.8	−45.3	2.8	226.2	N	207.9	87.1	3.5	2.2
MEAN					65	359.0	−46	2.6	46		3.3	−89.1	α_{95} = 2.6	k = 47.3

N = number of specimens per site; n = number of specimens included in the mean; Dec = declination; Inc = inclination; α_{95} and k are the statistical parameters of Fisher (1953); dm and dp define 95% confidence oval for the virtual geomagnetic poles; * not included in calculation of the geomagnetic pole.

width. In favorable outcrops a more detailed sampling was done including both margins as well as the center. Sample orientation was performed by both magnetic and sun compasses whenever possible.

Samples were cut into cylindrical specimens of 2.5 cm diameter and 2.2 cm height. At least three specimens from each dike were analyzed. Remanent magnetization was measured using a Digico spinner magnetometer. Samples were submitted to alternating field (a.f) demagnetizations, from fields of 2.5 or 5 mT up to 60 or 80 mT, in steps of 2.5 or 5 mT. For the majority of specimens a tumbling Molspin demagnetizer was used; a few samples were cleaned in a static Sapphire Instruments demagnetizer. Thermal demagnetizations were performed in a Schonstedt furnace in steps of 50°C up to 500°C and steps of 30°C up to 600°C. Remanences were completely erased below 600°C. Specimens from the same sample showed similar results for both cleaning methods.

Characteristic directions were determined by principal component analysis (Kirschvink, 1980) after selection of linear segments by visual inspection of Zijderveld (1967) plots. In general, the specimens showed a soft (low stability) component that could be cancelled by a.f. treatment at 20 mT. For greater fields, demagnetization paths trending toward the origin (Fig. 2) permitted the isolation of a single component of magnetization. Specimens showing a poorly defined demagnetization trajectory (MAD > 10°) were rejected. Specimens from 5 dikes gave inconsistent results.

Mean characteristic magnetization for each dike (Table 1; Fig. 3) was obtained by averaging directions from corresponding specimens, and statistical

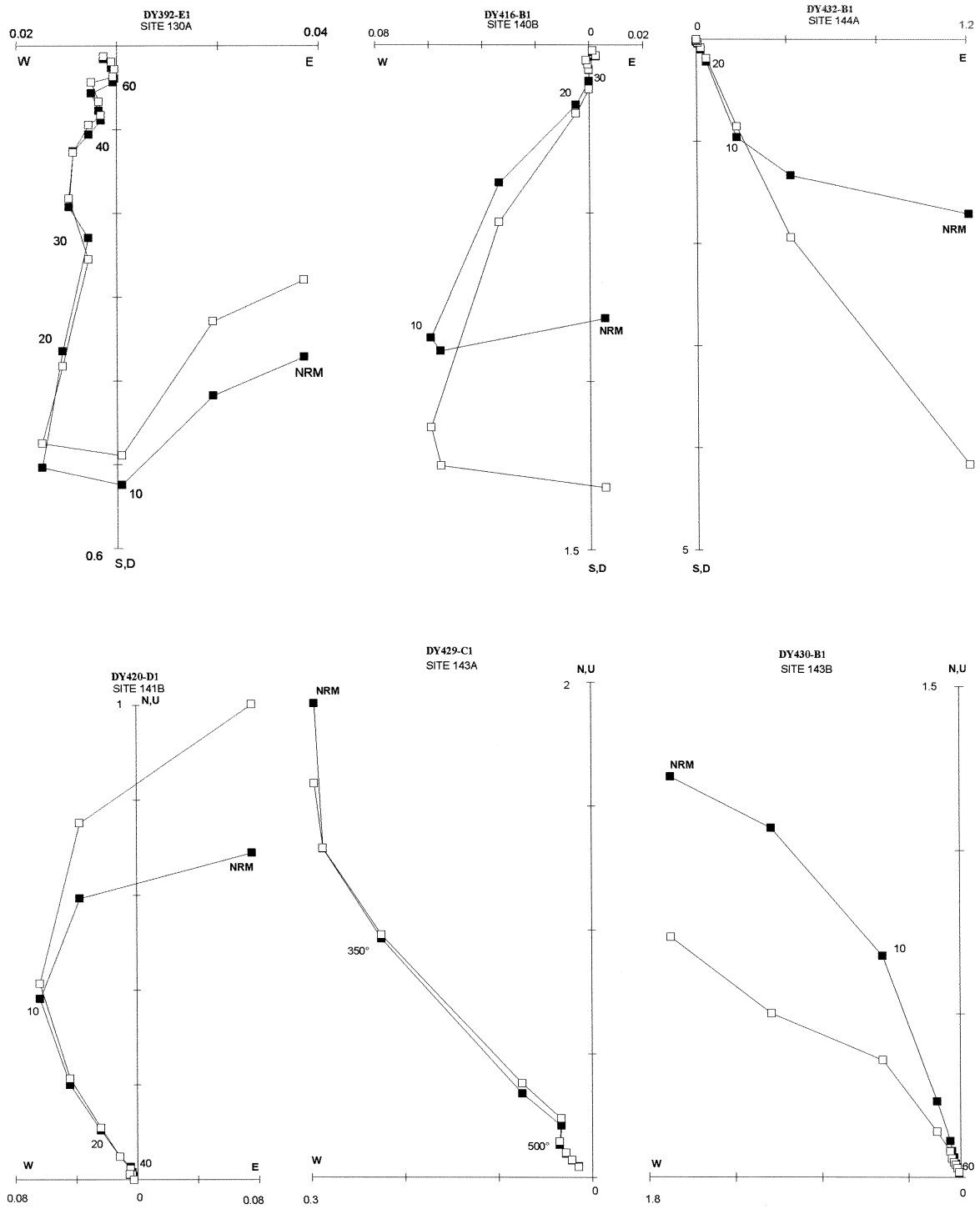


Fig. 2. Orthogonal diagrams for some specimens submitted to thermal (DY392-E1) and alternating field demagnetizations. Open symbols correspond to vertical plane, solid symbols to horizontal plane. Magnetization intensities in SI units, and a.f. fields in mT.

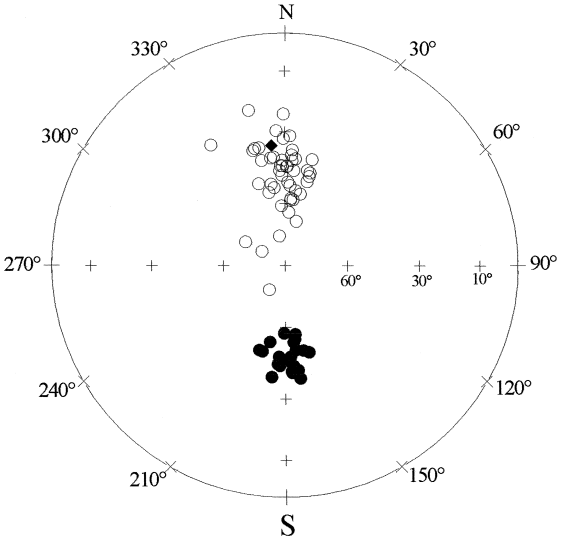


Fig. 3. Mean magnetization directions for 68 dykes plotted on Wulf stereogram. Open and solid symbols correspond to negative and positive inclinations, respectively. Diamond represents the local geomagnetic field.

parameters were calculated from Fisher (1953). Both normal and reversed polarities are recorded by FL dikes. It is worth noting that most of the NW dikes (except for site 179) are of normal polarities, and in Joaquina Beach (sites 130B and 130C in Table 1) they are seen cutting across one reversed polarity NE dike (site 130A). Where intrusive contacts between dikes could be observed (e.g., site 130), the older and harder reversed component is still preserved at a distance of about 3 m from the crossing with the younger NW normal polarity dike. Some normal polarity NE dikes (e.g., site 148 and 183) display soft secondary magnetizations of reversed polarity (Fig. 4) which were not destroyed by the present geomagnetic field induction nor any other recent physical–chemical process. Therefore, the normal polarity component carried by minerals with higher coercivities, cannot be attributed to present field overprinting. This means that although the normal polarity characteristic magnetization does not differ significantly from the present local geomagnetic field (Fig. 3), this remanence is probably of thermal origin and corresponds to the paleofield. Normal directions (Fig. 3) are slightly more scattered than the reversed ones due to the presence of some anomalous or intermediate (i.e., transitional polarity) directions, although the possible influence of incompletely removed secondary components must be considered.

The paleomagnetic pole (coded FL) calculated for the FL dikes is located at 3.3°E, 89.1°S ($N = 65$; $\alpha_{95} = 2.6°$; $k = 47$) and was computed as the mean of the virtual geomagnetic poles (VGPs) giving unit weight to each site. In this calculation data laying more than 30° apart from the mean were rejected (dikes 129A, 129B, and 154). The VGP angular dispersion for FL dikes ($S = 11.8°$) is in good agreement with the secular variation model proposed by McFadden and McElhinny (1984) for the period 45–110 Ma; for the period 110–195 Ma the angular dispersion is relatively low but their model is based on poorly constrained results. Normal (Dec = 359°; Inc = −46°; $N = 44$; $\alpha_{95} = 3.6°$; $k = 37$) and reversed (Dec = 179°; Inc = 47°; $N = 21$; $\alpha_{95} = 3.2°$; $k = 98$) mean directions are undistinguishable at the 95% confidence level when the McFadden and Lowes (1981) test is applied. This suggests that geomagnetic secular variation during both normal and reversed polarity intervals has been adequately sampled by the data set (Butler et al., 1991). Considering this and the distribution of the VGPs (Fig. 5) it is concluded that the geomagnetic secular variations were probably adequately sampled by the 73 studied dikes and their effects conveniently averaged from VGP mean allowing the calculation of a reference paleomagnetic pole.

4. Magnetic mineralogy

The main carrier of the remanence in FL dikes is (Ti-poor) magnetite as can be seen by Curie temperatures ranging from 560° to 580°C in typical thermomagnetic curves (Fig. 6) for samples of different polarities from the various sampling areas. Magnetite was observed in polished sections and is present mainly as microphenocrysts and groundmass occurrences in the majority of sections analyzed, although phenocrysts may be also seen. Trellis textures (with poorly developed lamellae), of composite and sandwich type (Haggerty, 1981) which indicate oxidation at high temperatures, are always found. Cruciform texture (Haggerty, 1981), which indicates rapid cooling, is present in some samples. Coercivities (H_c) of

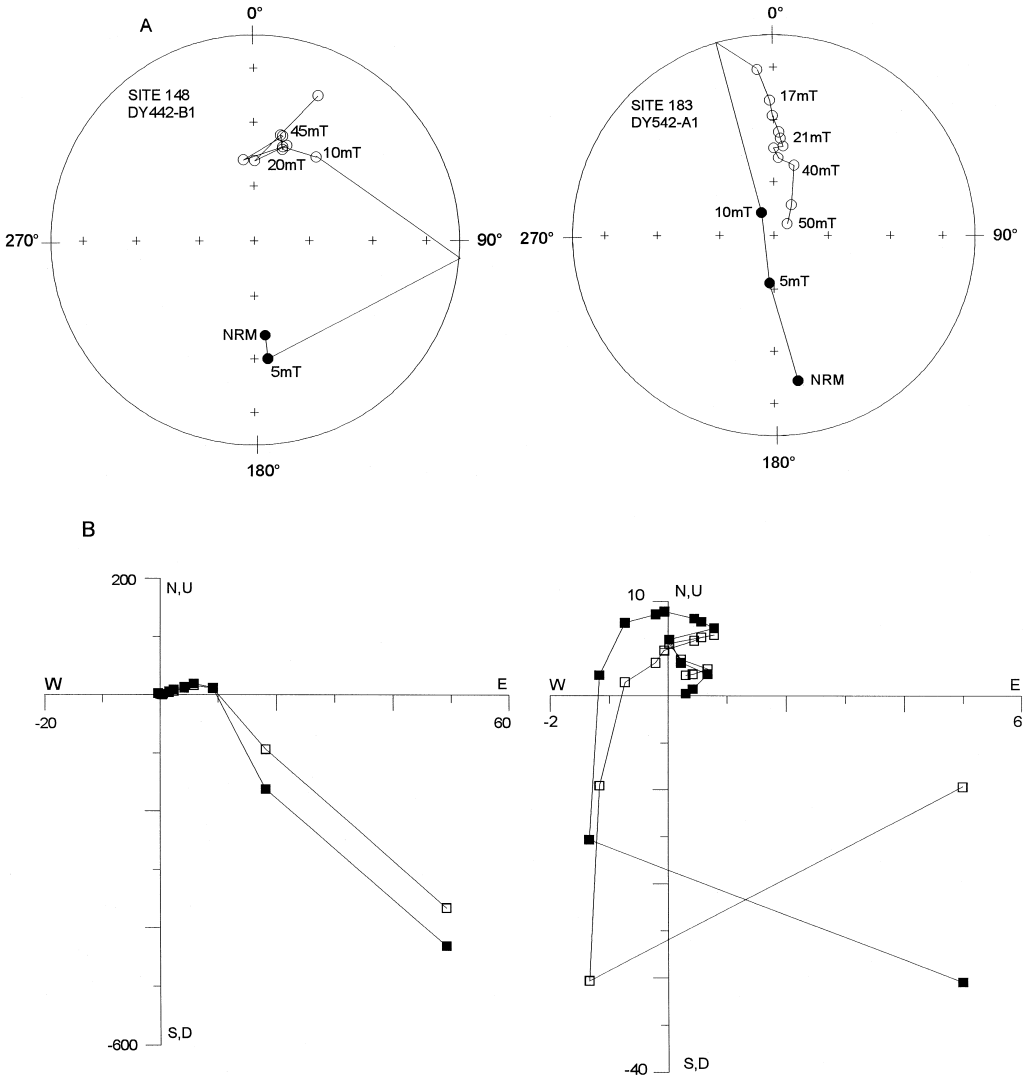


Fig. 4. (a) Demagnetization paths for specimens from two different sites showing a soft reversed magnetization component and a stable normal component. Open symbols are negative inclinations and solid symbols are positive inclinations. (b) Orthogonal plots for the same specimens; captions to diagrams same as in Fig. 2.

the magnetic grains in FL dikes calculated from hysteresis curves (Fig. 7) in a Molspin vibrating sample magnetometer vary from 6 mT to 21 mT; remanent coercive forces (H_{cr}) are in the range 18 mT to 45 mT. These values are consistent with the presence of magnetite or (Ti-poor) titanomagnetite in the dikes (Poornachandra Rao et al., 1992), in accordance with petrographic observations and thermomagnetic curves. These values indicate that the magnetic grains were able to preserve the original thermal remanent magnetizations.

5. $^{40}Ar/^{39}Ar$ dating

Plagioclase was separated from nine dikes, cleaned, irradiated, and analyzed using the methods and facilities described previously (Renne et al.,

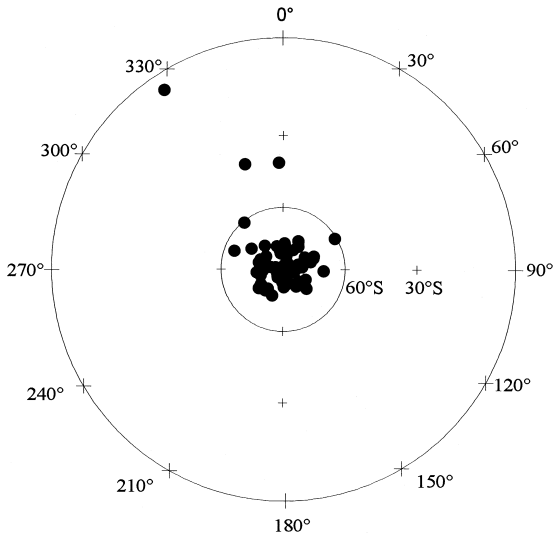


Fig. 5. Virtual geomagnetic poles plotted on Wulf stereogram. South projection.

1996a). Fish Canyon sanidine (FCs) was irradiated in five separate positions spanning the samples. Six analyses of each of the five monitors were used to calculate a weighted mean J value for each position, and the arithmetic mean and standard deviation of these five J values (0.003328 ± 0.000011) was used to calculate apparent ages relative to an age of 27.84 Ma for FCs. The true age of FCs is likely to be ~1% older than this (e.g., Renne et al., 1998) but the previous value is used for consistency with copious previous $^{40}Ar/^{39}Ar$ data (Renne et al., 1992a,b, 1996a,b, 1993; Erlank et al., 1984;) for early Cretaceous magmatism in South America and Africa, which were based on 27.84 Ma for FCs or the analogous age of 520.4 Ma for Mmhb-1.

Apparent age spectra (Fig. 8) for the plagioclase samples were markedly discordant, with virtually all displaying higher apparent ages with more extensive outgassing. Apparent Ca/K values also increase with increasing extraction temperatures, consistent with the petrographic observation of normal zoning (i.e., more calcic cores) of plagioclase in thin section. Many spectra also show high apparent ages at lower temperatures, and several show saddle shaped spectra reminiscent of the effects of excess ^{40}Ar (e.g., Lanphere and Dalrymple, 1976). The presence of excess ^{40}Ar is consistent with the fact that the dikes intrude 600 Ma, high-K granitoid basement rocks.

Apparent age spectra for five samples (FL-13, -28, -30, -81, -82, corresponding to paleomagnetic sites 125A, 130A, 130C, 150, and 151, respectively) displayed plateau-like features according to the criteria of Renne et al. (1996a) except that the plateaus occupy only 40–60% of the total ^{39}Ar released. These plateau ages range from 119.0 ± 0.9 Ma to 128.3 ± 0.5 Ma. Most of the plateaus involve steps yielding younger apparent ages than most of the rest

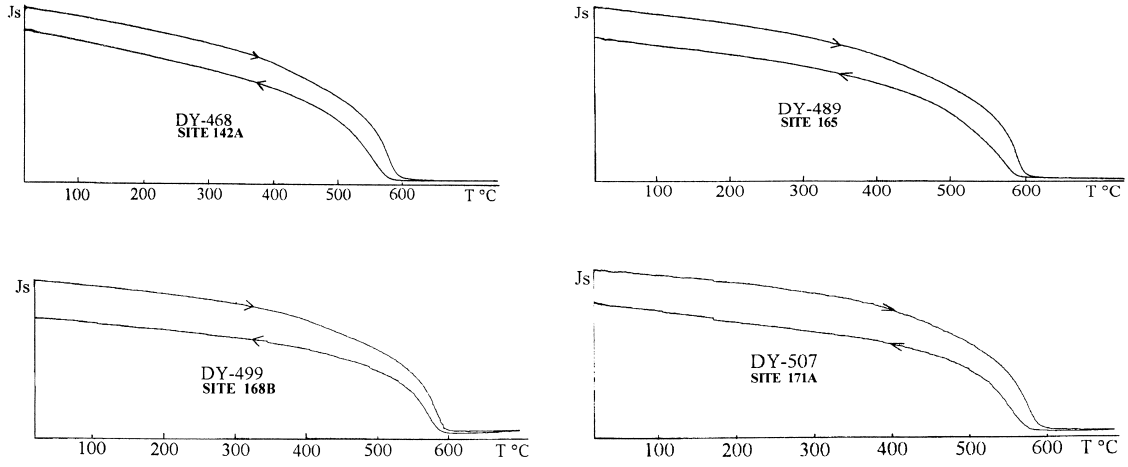


Fig. 6. Examples of the thermomagnetic curves obtained for the FL dykes. Heating rate of 50°C min^{-1}; J_s is saturation magnetization in arbitrary units.

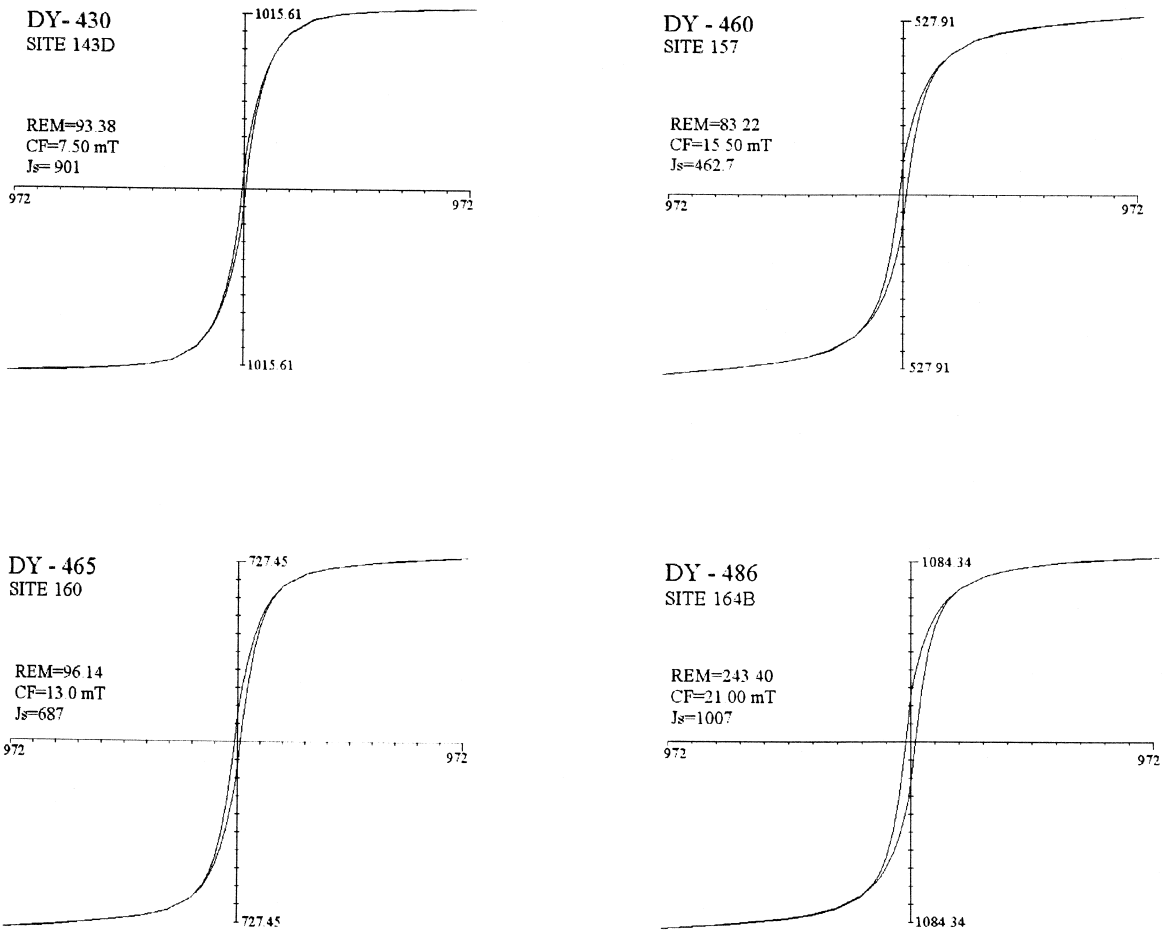


Fig. 7. Hysteresis curves for some studied samples. REM is the remanent magnetization in μAm^2; CF is the coercive force, and J_s is the saturation magnetization in μAm^2.

of the spectrum, but the plateau age for FL-28 (PM-130A), 128.3 ± 0.5 Ma, involves the oldest steps in a hump-shaped spectrum, showing evidence of $^{40}Ar*$ loss in low temperature steps and slight Ar-recoil effects at high temperature steps.

Isotope correlation diagrams of $^{36}Ar/^{40}Ar$ vs. $^{39}Ar/^{40}Ar$ show excess scatter in most cases, indicating that more than two components of Ar are present and that in most samples the various components have not been resolved into binary systems by incremental heating. Three samples yielded reasonable isochrons (Fig. 9) in that they had MSWD values less than 2.5; two of these (FL-81 and FL-82; PM-150 and -151) were also samples that yielded apparent age plateaus. In both of these cases the isochron ages (127.3 ± 0.5 and 125.8 ± 0.5 Ma, respectively) for plateau steps are slightly younger than plateau ages (128.2 ± 0.4 and 126.2 ± 0.4 Ma, respectively), though the ages calculated in both ways are indistinguishable within analytical errors. A third isochron was obtained from sample FL-90 (PM-158), which did not yield a plateau. The isochron was fit to data from the second to eighth steps, occupying the low-temperature side of the saddle, and corresponds to an age of 121.5 ± 0.5 Ma.

We interpret the isochron ages for dikes FL-81, -82, and -90 (PM-150, -151, and 158), as well as the three additional plateau ages for dikes FL-13, -28, and -30 (PM-125A, -130A, and -130C), to represent the most likely ages for the intrusions of these dikes.

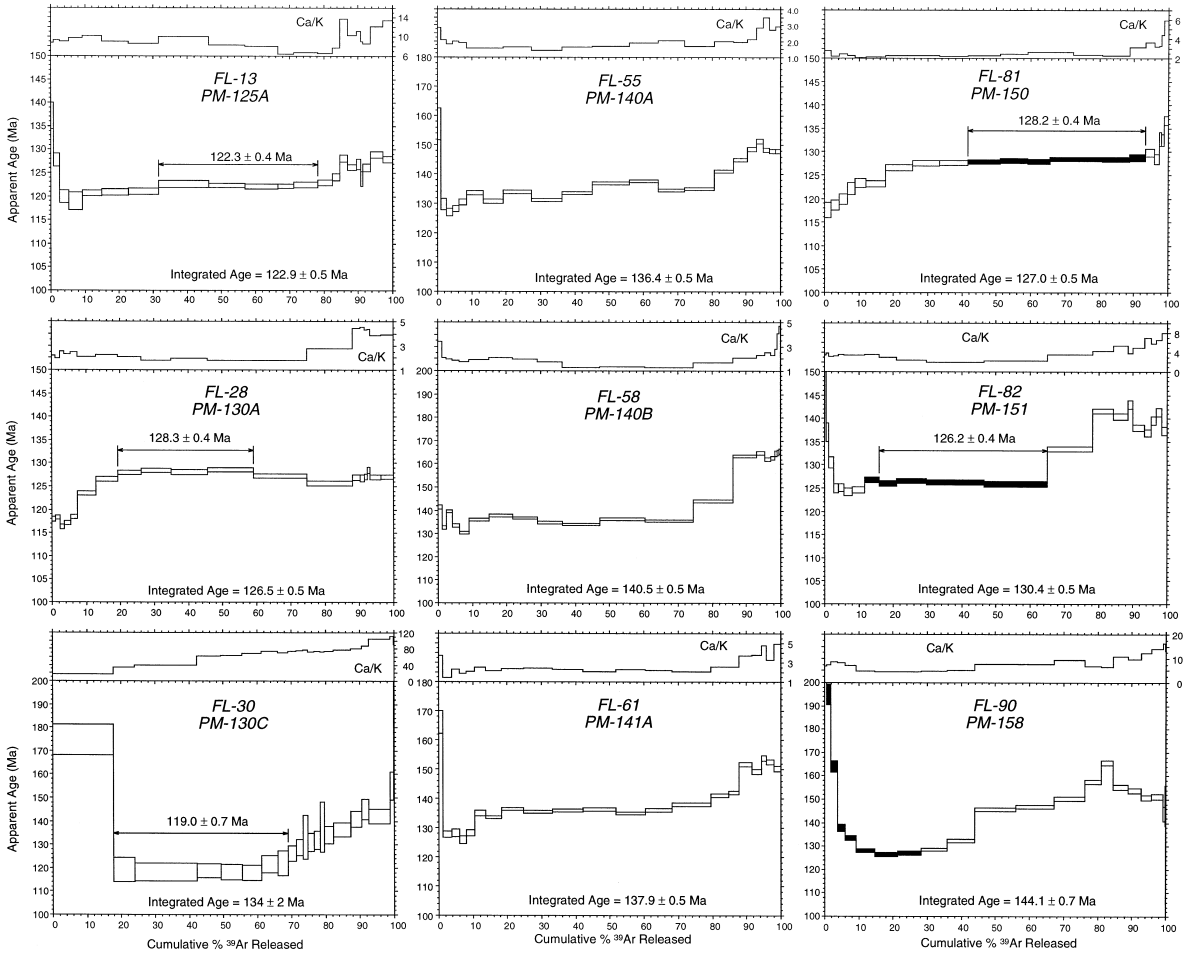


Fig. 8. Apparent age and Ca/K spectra for plagioclase separates from 9 dikes. Number preceded by 'FL' is the geochronology sample number; corresponding paleomagnetic site number is preceded by 'PM'. Individual step ages shown with 95% confidence limits excluding systematic errors. Plateau ages (where defined) are indicated. Portions of spectra used to calculate isochrons (Fig. 9) are shown in black (FL-81, -82, -90; PM-150, -151, -158).

The plateau age (128.3 ± 0.5 Ma) for FL-28 (PM-130A) is considered least reliable of the plateau ages because of the anomalous shape of the spectrum and because it contains $< 40\%$ of the total ^{39}Ar released from this sample.

Accepting these results as crystallization ages indicates that the Florianópolis dikes were intruded over the range of about 119 to 128 Ma. Though it is possible that the ~ 9 m.yr. spread in these ages is partly an artifact of excess argon effects, there is no reason to reject a priori such a long duration of dike injection activity. In the one case where cross-cutting relations can be used to check the results, the younger NW-trending dike FL-30 (PM-130C; 119.0 ± 0.9 Ma plateau age) cuts across the NE-trending dike FL-28 (PM-130A; 128.3 ± 0.5 Ma plateau age), thus providing a qualitative validation of the relative age.

Though the $^{40}Ar/^{39}Ar$ data appear to define two modes at ~ 119–122 Ma and ~ 126–128 Ma, the data are insufficient to determine whether dike intrusion occurred continuously or episodically. In this connection, it is important to note that a one to one correspondence does not exist between dike trend and age, since two NE-trending dikes (FL-13 and -90; PM-125A, -158) also yielded ages nearly as young as the youngest NW-trending one.

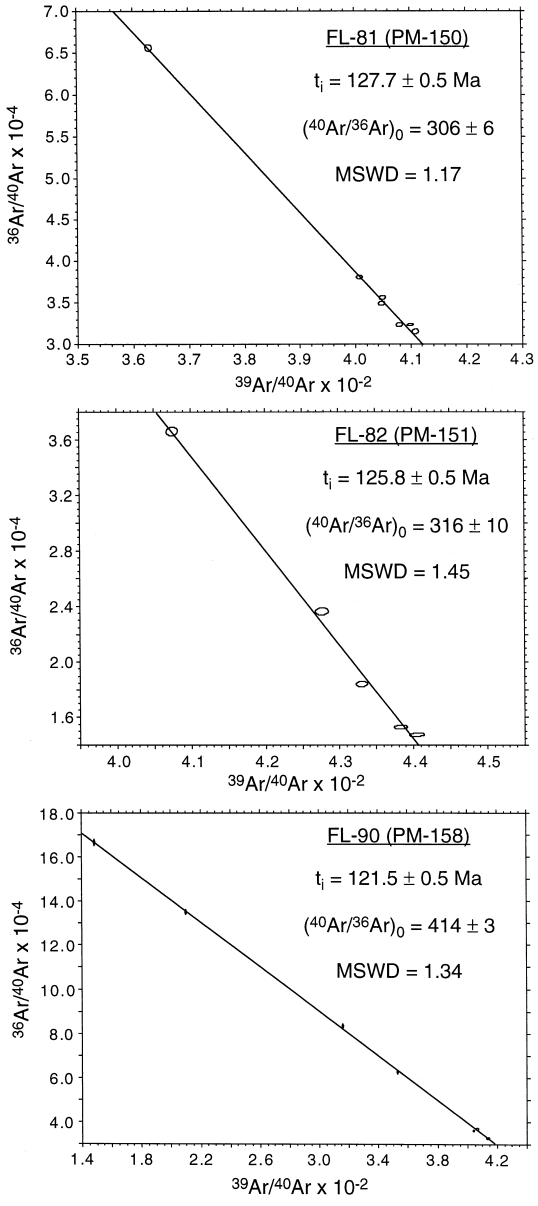


Fig. 9. Isochrons for three samples meeting criteria discussed in text. Diagrams are restricted to the portion of isotope correlation plot containing the data used for isochron calculation. t_i = isochron age; $(^{40}Ar/^{36}Ar)_0$ = trapped argon isotope ratio; MSWD = mean squared weighted deviates.

It is noteworthy that none of the ages determined for the dikes coincide with ages determined for the nearby Paraná flood volcanic rocks (Renne et al., 1992a; Turner et al., 1994) or the Etendeka volcanics (Renne et al., 1996b) which were nearby prior to continental drift. The possibility that some of the dikes are this old cannot be discounted, but the implication nonetheless is that feeders for the southern Paraná-Etendeka volcanics lay elsewhere.

6. Discussion

The Florianópolis dikes yield $^{40}Ar/^{39}Ar$ dates consistent with their intrusion during development of the East Brazilian continental margin, a passive margin formed as a consequence of crustal breakup in the Late Aptian (Chang et al., 1992). There are few paleomagnetic poles related to this tectonic event, and no detailed Mesozoic APWP for South America can be traced so far. The most reliable (large number os sites and well constrained ages) paleomagnetic poles are of Early Cretaceous ages (Table 2), and a good mean Late Cretaceous (mainly 70–90 Ma) was calculated by Montes-Lauar et al. (1995). The only available pole for mid-Cretaceous is the one based on the Cerro Barcino sediments from central Patagonia, western Argentina (Somoza, 1994), of Aptian age. The paleomagnetic pole calculated for the Florianópolis dikes (FL pole) is compared (Fig. 10) to the above mentioned Cretaceous poles for South America (Table 2). The FL pole is not distinguishable from the CB pole at the 95% confidence level. However, as indicated by its large confidence circle, the CB pole may lack the required precision and may be actually younger than Florianópolis dikes.

The 95% confidence circle of the FL pole does not intersect the circles of confidence of the paleomagnetic poles from the Serra Geral volcanics, and the Ponta Grossa dikes for which a main age of 130.5 Ma has been determined although minor activity in the 121–126 Ma interval was also observed (Renne et al., 1996a). However, the PG pole may not represent a reference pole as it differs statistically from the SG pole, although their ages differ by 3 Ma. Raposo and Ernesto (1995) interpreted the 6.9° angular discrepancy between the SG and PG poles as the displacement of the western Gondwana block, on the basis of a time interval of about 10 Ma. However, as pointed out by Renne et al. (1996a), this angular discordance may be partly due to intraplate

Table 2
Selected cretaceous paleomagnetic poles for South America

	$N/(n)$	Paleomagnetic poles				Code	Refs.
		Long. (°E)	Lat. (°S)	α_{95} (°)	k		
Late Cretaceous	(5)	349.1	81.0	3.5	492	LK	Montes-Lauar et al., 1995
Cerro Barcino Formation	11	180.8	84.9	5.6	66	CB	Somoza, 1994
Florianópolis Dikes	65	3.3	89.1	2.6	47	FL	This paper
Florianópolis Dikes—Normal	44	7.6	89.3	3.5	39		This paper
Florianópolis Dikes—Reversed	21	358.6	88.7	3.5	82		This paper
Ponta Grossa Swarm	115	30.3	82.4	2.0	44	PG	Raposo and Ernesto, 1995
Central Alkaline Province	75	62.3	85.4	3.1	29	CAP	Ernesto et al., 1996
Paraná Basin Volcanics	339	90.1	84.3	1.2	40	SG	Raposo and Ernesto, 1995

N = number of sites; (n) = number of poles; α_{95} and k are the statistical parameters of Fisher (1953).

deformation, since plate velocities as high as 40 cm/yr are required since the age difference between the SG and PG poles is only ~2–3 Ma. On the other hand, the paleomagnetic pole for the Central Alkaline Province (CAP) in Paraguay (Ernesto et al., 1996), which probably refers to the same time interval (127–130 Ma) as the Ponta Grossa dikes based on K/Ar dating (Velázquez et al., 1992) is closer to the SG pole, describing an arc of ~2.7°, and therefore implying more reasonable velocities. This observation reinforces the hypothesis of a tectonic influence on the PG pole, although the subject needs further investigation and will be treated elsewhere.

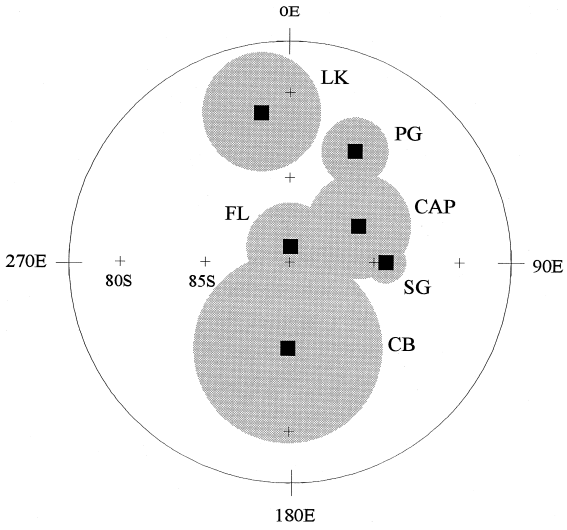


Fig. 10. Plot of the Florianópolis (FL) dikes paleomagnetic pole (this paper) along with the paleomagnetic poles for the Paraná Basin volcanics (SG) and the Ponta Grossa dikes (PG) (Raposo and Ernesto, 1995), the alkaline rocks from central-eastern Paraguay (Ernesto et al., 1996), the mid-Cretaceous Cerro Barcino Formation (CB) (Somoza, 1994), and the mean Late Cretaceous pole (Kl) (Montes-Lauar et al., 1995). Shaded areas correspond to the 95% confidence circles.

The FL pole is not statistically different from the CAP pole, at a 95% confidence level, as expected if the FL VGPs are dominated by dikes in the 126–128 Ma range. However, a significant proportion of the dike ages appear to be 119–122 Ma, and it appears likely that the FL pole represents a time-averaged pole with mean age of about 124 ± 4 Ma. The clustering of VGPs (Fig. 5) suggests that little plate motion occurred within the time span of intrusion of the sampled dikes, providing some justification for combining the data in this fashion.

It is believed by some workers that rifting in the nascient South Atlantic evolved from south to north, reaching latitudes of about 38°S at 130 Ma (Nürnberg and Müller, 1991). According to Chang et al. (1992) north of 33°S seafloor spreading commenced around anomaly M4 (126.5 Ma). Therefore, the Florianópolis dikes are probably the late record of the crustal extension in the area. Northwards, in the Santos and Campos (~20°S) offshore basins, significant volcanism postdating the Serra Geral basalts and of Neocomian age (Mizusaki et al., 1992) is probably correlative with the Florianópolis dikes. The Florianopólis dikes are likely equivalent to the Horingbaai dikes and sills of Namibia, which have yielded $^{40}Ar/^{39}Ar$ plateau ages of ~125 Ma (Renne et al., 1996b); other basaltic dikes in north-coastal Namibia also yield similar results (P.R. Renne, un-

published data). These dikes and sills appear to reflect coast-parallel extension along the once adjacent margin of southwestern Africa.

Using Debiche and Watson (1995) method, an anticlockwise rotation of $3.3° \pm 0.4°$, and a southward latitudinal displacement of $4.9° \pm 1.8°$ is calculated for the South American plate, within the ~ 8 m.yr. between the Serra Geral event and the mean age of emplacement of the FL dikes. The arc distance of about 5.7° between the FL and SG poles implies a minimum velocity of ~ 5.5 cm/yr over this time interval. The above rotation is not consistent with the current model of a rift propagating from south to north causing a clockwise rotation of South American plate (e.g., Nürnberg and Müller, 1991), and suggests that the dikes were emplaced before the effective formation of oceanic crust at this latitude.

The difference between FL and SG or CAP poles could also be explained by a local rotation of Santa Catarina Island, perhaps during rifting. Although this hypothesis cannot be entirely excluded based on the present data, the available $^{40}Ar/^{39}Ar$ ages indicate that the pole should fall between the SG and Late Cretaceous poles on the South America APWP. Thus it seems more likely that the discordance is due to plate motion than local rotation. Comparison with African paleomagnetic poles of same age would help to test this hypothesis, however reliable poles of this age are not available yet. A clockwise rotation of about 6° is detected between the FL pole and the mean Late Cretaceous ($\sim 90–70$ Ma) pole for South America (Montes-Lauar et al., 1995) consistent with the expected rotation considering that the continental crusts of Northeastern Brazil and Africa did not separate before Late Albian to Early Cenomanian times (Nürnberg and Müller, 1991).

7. Summary and conclusions

The $^{40}Ar/^{39}Ar$ and paleomagnetic data presented in this paper allow the following main conclusions.

(1) Ages of the Florianópolis dikes are dominantly 119–128 Ma. Present data define two modes at $\sim 119–122$ Ma and $\sim 126–128$ Ma, although they are probably insufficient to determine whether dike intrusion occurred continuously or episodically.

(2) The FL paleomagnetic pole is at 3.3°E, 89.1°S ($N = 65$; $\alpha_{95} = 2.6°$; $k = 47$), and differs from other South American poles (Raposo and Ernesto, 1995; Ernesto et al., 1996) spanning ages of $\sim 127–130$ Ma, consistent with the conclusion that the FL data set is dominated by younger dikes. The FL pole represents a reference pole for which a mean age of 124 ± 4 Ma is proposed; this age is the mean of the six $^{40}Ar/^{39}Ar$ dates determined for FL dikes, and the uncertainty encompasses the range of ages inferred.

(3) The FL pole differs at the 95% confidence level from the Serra Geral pole (Ernesto et al., 1990; Raposo and Ernesto, 1995) implying an anticlockwise rotation of $3.3° \pm 0.4°$ and a southward migration of $4.9° \pm 1.8°$ for the time span between the main volcanic activity in the Paraná Basin (133–132 Ma; Renne et al., 1992a,b) and the emplacement of the Florianópolis dikes.

(4) A clockwise rotation for South America predicted by current South Atlantic opening models is inferred from the pole path between FL and the mean Late Cretaceous (Montes-Lauar et al., 1995) paleomagnetic poles. Ages of 128–119 Ma are coeval with development of the East Brazilian continental margin, thus it is concluded that the FL dikes were emplaced before the effective formation of oceanic crust at this latitude.

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